

Chapter 3 — Top 10 Exam-Ready Questions & Answers

Q1. What is the difference between functional and non-functional requirements in software engineering, and where are they documented?

বাংলা অর্থ

Functional Requirement বলে software **কী কী কাজ করবে**।

যেমন: User login করতে পারবে।

Non-functional Requirement বলে software **কীভাবে/কত ভালোভাবে কাজ করবে**।

যেমন: System একসাথে ২০ লাখ user handle করতে পারবে।

Exam-Ready Answer

In Software Engineering, functional requirements describe what the system should do, while non-functional requirements describe the quality and constraints of the system.

In frameworks like the Rational Unified Process (RUP), functional requirements are usually captured in Use Case stories.

Non-functional requirements include quality factors such as usability, capacity, security, reliability, and maintainability. They are typically documented in the Software Requirements Specification (SRS).

সহজভাবে মনে রাখো

Functional = What the system does

Non-functional = How well the system does it

Q2. Why is there a widespread need for comprehensive software quality requirements?

বাংলা অর্থ

অনেক সময় requirements document-এ শুধু software **কী কাজ করবে** সেটা লেখা থাকে। কিন্তু software কতটা **usable, reliable, secure, maintainable** হবে—এসব লেখা থাকে না।

তাই comprehensive quality requirements দরকার।

Exam-Ready Answer

The need for improving software requirements documents is widespread because they frequently lack essential quality factors such as usability, reusability, and maintainability.

To address this, the software industry groups related attributes into comprehensive quality factors, also called non-functional requirements.

However, different projects require different levels of emphasis on quality factors. For example, scalability may be more important for one project, while reliability may be more important for another.

সহজভাবে

সব project-এর quality requirement একই হয় না।

Banking system-এ security বেশি গুরুত্বপূর্ণ, আর real-time control system-এ safety বেশি গুরুত্বপূর্ণ হতে পারে।

Q3. Outline McCall's Software Quality Factors model and its three primary categories.

বাংলা অর্থ

McCall software-এর quality মাপার জন্য ১১টি quality factor দিয়েছেন এবং এগুলোকে ৩টি category-তে ভাগ করেছেন।

Exam-Ready Answer

McCall's model (1977) categorizes 11 software quality factors into three main categories.

1. Product Operation Factors: These assess how well the software operates. They include correctness, reliability, efficiency, integrity, and usability.

2. Product Revision Factors: These evaluate how easily the software can be changed, maintained, and tested. They include maintainability, flexibility, and testability.

3. Product Transition Factors: These evaluate how easily the software can be moved to different platforms and interact with other systems. They include portability, reusability, and interoperability.

সহজে মনে রাখো

Operation → এখন কেমন চলছে?

Revision → পরে পরিবর্তন করা কত সহজ?

Transition → অন্য জায়গায় নেওয়া/ব্যবহার করা কত সহজ?

3 Category + Factors

Category

Factors

Operation Correctness, Reliability, Efficiency, Integrity, Usability

Revision Maintainability, Flexibility, Testability

Transition Portability, Reusability, Interoperability

5 + 3 + 3 = 11 factors

Q4. Discuss the specific factors included under McCall's "Product Operation" category.

বাংলা অর্থ

Product Operation বলতে বোঝায়:

Software এখন production-এ কত ভালোভাবে কাজ করছে?

এর মধ্যে ৫টি factor আছে।

Exam-Ready Answer

The Product Operation category consists of five factors:

- 1. Correctness:** Measures whether the software meets its specified requirements and produces accurate and complete outputs.
- 2. Reliability:** Measures the ability of the software to provide service without failure and within an acceptable failure rate or downtime.
- 3. Efficiency:** Measures the hardware and system resources required to perform the software functions, such as processing power, memory, storage, and communication speed.
- 4. Integrity:** Deals with protecting the system and data from unauthorized access.
- 5. Usability:** Measures the ease of learning, operating, and using the software, including the effort required to train users.

বাংলা অর্থ

Correctness → Requirement অনুযায়ী সঠিক output দিচ্ছে কি না।

Reliability → Failure ছাড়া কত ভালোভাবে service দিতে পারে।

Efficiency → CPU, RAM, storage ইত্যাদি কত efficiently ব্যবহার করছে।

Integrity → Unauthorized access থেকে system/data নিরাপদ রাখা।

Usability → User-এর জন্য software কত সহজে ব্যবহারযোগ্য।

Q5. Describe the components of "Product Revision Software Factors" in McCall's model.

বাংলা অর্থ

Software release হওয়ার পরবর্তীতে যখন bug fix, modification বা নতুন feature যোগ করা হবে, তখন Product Revision factors গুরুত্বপূর্ণ।

এখানে ৩টি factor:

Exam-Ready Answer

Product Revision Factors deal with requirements related to software maintenance activities.

1. Maintainability: Measures the effort required to identify, correct, and verify software failures. Good modularity and documentation improve maintainability.

2. Flexibility: Measures the effort required to modify or adapt the software for different customers, requirements, or environments.

3. Testability: Measures how easily the software can be tested. Logs, predefined intermediate results, and self-diagnostic features can improve testability.

বাংলা অর্থ

Maintainability → Bug খুঁজে fix করা কত সহজ।

Flexibility → নতুন requirement বা customer-এর জন্য software পরিবর্তন করা কত সহজ।

Testability → Software সহজে test করা যায় কি না।

☐ Shortcut:

MFT → Maintainability, Flexibility, Testability

Q6. What are Product Transition factors, and why is Reusability highly significant?

বাংলা অর্থ

Product Transition বলতে বোঝায় software-কে অন্য platform/environment-এ নেওয়া, অন্য system-এর সাথে connect করা এবং software-এর অংশ আবার ব্যবহার করা।

৩টি factor:

Exam-Ready Answer

Product Transition Factors evaluate whether an application can be moved to different environments or interface with other systems.

- 1. Portability:** Measures the ability to move software to different hardware, operating systems, or environments.
- 2. Interoperability:** Measures the ability of an application to interact with other existing systems or databases.
- 3. Reusability:** Measures the ability to reuse software components in developing new applications. It can significantly reduce development time and cost and improve software quality.

বাংলা অর্থ

Portability → অন্য OS/platform-এ software চালানো।

Interoperability → অন্য system/database-এর সাথে communicate করা।

Reusability → পুরোনো code/component নতুন project-এ আবার ব্যবহার করা।

Reusability কেন গুরুত্বপূর্ণ?

একবার ভালো একটি component তৈরি করলে সেটা বারবার ব্যবহার করা যায়।

☞ সময় কমে

☞ খরচ কমে

🔗 Development দ্রুত হয়

🔗 Quality improve হয়

Q7. What are the alternative quality factors introduced by Evans and Marciniak?

বাংলা অর্থ

McCall-এর model-এর পাশাপাশি **Evans and Marciniak** আরও দুইটি quality factor প্রস্তাব করেছেন:

1. **Verifiability**
2. **Expandability**

Exam-Ready Answer

Evans and Marciniak introduced two additional quality factors: Verifiability and Expandability.

1. Verifiability: Refers to design and programming features that allow efficient verification of the software. It focuses on aspects such as modularity, simplicity, coupling, cohesion, and programming standards.

2. Expandability: Refers to the ability of software to be extended or scaled to meet new requirements. It is closely related to McCall's flexibility factor.

বাংলা অর্থ

Verifiability → Software-এর design/code সহজে verify বা check করা যায় কি না।

এখানে দেখা হয়:

- Modularity
- Simplicity
- Coupling
- Cohesion
- Coding guidelines

Expandability → ভবিষ্যতে নতুন feature বা বড় load যোগ করা কত সহজ।

🔗 এটাকে **Scalability + Extensibility** হিসেবেও মনে রাখতে পারো।

Q8. Describe the alternative quality factors proposed by Deutsch and Willis.

বাংলা অর্থ

Deutsch and Willis ৩টি quality factor দিয়েছেন:

1. **Safety**
2. **Manageability**
3. **Survivability**

Exam-Ready Answer

Deutsch and Willis proposed three alternative quality factors: Safety, Manageability, and Survivability.

1. Safety: Addresses conditions that could cause the equipment or application to fail or become unsafe. It is especially important for real-time control software.

2. Manageability: Refers to administrative tools used to control software versions, configurations, and changes.

3. Survivability: Refers to the ability of software to continue providing service despite failures. It is commonly measured using MTBF (Mean Time Between Failures) and MTTR (Mean Time To Recover).

বাংলা অর্থ

Safety → Software বিপজ্জনক পরিস্থিতিতে নিরাপদভাবে কাজ করবে।

Manageability → Version, configuration এবং changes সহজে manage/control করা।

Survivability → Failure হলেও software service চালু রাখতে পারা এবং দ্রুত recover করা।

☐ **Shortcut:**

S-M-S

Safety → **Safe**

Manageability → **Manage**

Survivability → **Survive**

Q9. Differentiate between Reliability and Efficiency within McCall's Product Operation Factors.

বাংলা অর্থ

এখানে **Reliability** এবং **Efficiency**-এর পার্থক্য জানতে চাওয়া হয়েছে।

Exam-Ready Answer

Reliability refers to the ability of software to provide service without failure. It is concerned with failure rates and downtime.

Efficiency, on the other hand, refers to the hardware and system resources required to perform software functions. It includes processing power, memory, storage, and communication speed.

Therefore, reliability focuses on **failure-free service**, while efficiency focuses on **resource usage and performance**.

খুব সহজে:

Reliability = কতক্ষণ ঠিকভাবে চলবে?

Efficiency = কত কম resource দিয়ে কাজ করবে?

Example:

ধরো Banking System:

- System ৩০ দিনেও crash করল না → Reliability
- System কম RAM/CPU ব্যবহার করে দ্রুত কাজ করল → Efficiency

Q10. Who is responsible for defining these quality factors, and where are they documented in a modern development lifecycle?

বাংলা অর্থ

Quality factor শুধু developer-এর দায়িত্ব নয় এবং শুধু client-এর দায়িত্বও নয়।

Client + Developer উভয়েরই quality requirement নিয়ে কাজ করতে হয়।

Exam-Ready Answer

Both developers and clients are responsible for defining and considering software quality factors, although the importance of each factor may vary depending on the project.

The nature of the application determines which quality factors require greater emphasis. For example, safety is more important in real-time control systems, while security may be critical in banking systems.

In processes such as the Rational Unified Process (RUP), these non-functional requirements are explicitly documented in the Software Requirements Specification (SRS).

বাংলা অর্থ

Client এবং developer মিলে ঠিক করবে:

- Software কতটা secure হবে?
- কত fast হবে?
- কতজন user handle করবে?
- কত reliable হবে?
- কত সহজে maintain করা যাবে?

এসব requirement সাধারণত **SRS document-এ** লেখা হয়।

★ Bonus Short Notes

1. Non-functional Requirements

বাংলা অর্থ

Non-functional requirement বলে software **কী কাজ করবে সেটা নয়**; বরং software **কত ভালোভাবে কাজ করবে সেটা** বোঝায়।

Exam-Ready Definition

Non-functional requirements define the quality attributes and constraints of a software system rather than its functional behavior. Examples include usability, security, scalability, reliability, and maintainability.

Example:

“User login করতে পারবে” → Functional

“Login process must complete within 2 seconds” → Non-functional

2. Verifiability

বাংলা অর্থ

Software-এর **design** এবং **code** কত সহজে **verify/check** করা যায়।

এখানে code-এর:

- Modularity
- Simplicity
- Coupling
- Cohesion

ইত্যাদি দেখা হয়।

Exam-Ready Definition

Verifiability is a quality factor that measures how efficiently the design and programming of a software system can be verified.

3. Survivability

বাংলা অর্থ

Software failure হওয়ার পরও **service** চালু রাখতে বা দ্রুত **recover** করতে পারার ক্ষমতা।

এখানে:

MTBF = Failure-এর মাঝখানে গড় সময়

MTTR = Failure হওয়ার পর **recover** করতে গড় সময়

Exam-Ready Definition

Survivability is the ability of software to continue or quickly restore its service after failures. It is commonly measured using MTBF and MTTR.

4. SRS — Software Requirements Specification

বাংলা অর্থ

SRS হলো software-এর **formal requirements document**, যেখানে system-এর requirements বিস্তারিতভাবে লেখা থাকে।

বিশেষ করে **non-functional requirements/quality factors** এখানে document করা হয়।

Exam-Ready Definition

SRS (Software Requirements Specification) is a formal document that specifies the requirements of a software system, including its functional and non-functional requirements.

এক নজরে পুরো Chapter

বিষয়

সহজ অর্থ

Functional	System কী কাজ করবে
Non-functional	System কত ভালোভাবে কাজ করবে
McCall	11 Quality Factors
Product Operation	এখন software কেমন চলছে
Product Revision	Software পরিবর্তন/maintain করা কত সহজ
Product Transition	অন্য platform/system-এ নেওয়া কত সহজ
Evans & Marciniak	Verifiability + Expandability
Deutsch & Willis	Safety + Manageability + Survivability
Reliability	Failure ছাড়া চলার ক্ষমতা
Efficiency	Resource efficiently ব্যবহার
SRS	Requirements-এর formal document

☐ McCall-এর 11 Factors মনে রাখার সবচেয়ে সহজ উপায়

Operation (5):

C R E I U

→ Correctness, Reliability, Efficiency, Integrity, Usability

Revision (3):

M F T

→ Maintainability, Flexibility, Testability

Transition (3):

P R I

→ Portability, Reusability, Interoperability

Total = 5 + 3 + 3 = 11 ✓